

```
!pip install langchain_community langchainhub chromadb langchain langchain-openai
```

```
Requirement already satisfied: google-auth==1.0.1 in /usr/local/lib/python3.11/dist-packages (from kubernetes>=28.1.0->chromadb)
Requirement already satisfied: websocket-client!=0.40.0,!0.41.*,!=0.42.*,>=0.32.0 in /usr/local/lib/python3.11/dist-packages (from kubernetes>=28.1.0->chromadb)
Requirement already satisfied: requests-oauthlib in /usr/local/lib/python3.11/dist-packages (from kubernetes>=28.1.0->chromadb)
Requirement already satisfied: oauthlib==3.2.2 in /usr/local/lib/python3.11/dist-packages (from kubernetes>=28.1.0->chromadb)
Requirement already satisfied: urllib3>=1.24.2 in /usr/local/lib/python3.11/dist-packages (from kubernetes>=28.1.0->chromadb)
Requirement already satisfied: durationpy==0.7 in /usr/local/lib/python3.11/dist-packages (from kubernetes>=28.1.0->chromadb)
Requirement already satisfied: jsonpatch<2.0,>=1.33 in /usr/local/lib/python3.11/dist-packages (from langchain-core<1.6.0,>=0.1.0->langchain)
Requirement already satisfied: requests-toolbelt<2.0.0,>=1.0.0 in /usr/local/lib/python3.11/dist-packages (from langsmith>=0.1.17->langchain)
Requirement already satisfied: zstandard<0.24.0,>=0.23.0 in /usr/local/lib/python3.11/dist-packages (from langsmith>=0.1.17->langchain)
Requirement already satisfied: coloredlogs in /usr/local/lib/python3.11/dist-packages (from onnxruntime==1.14.1->chromadb)
Requirement already satisfied: flatbuffers in /usr/local/lib/python3.11/dist-packages (from onnxruntime==1.14.1->chromadb)
Requirement already satisfied: protobuf in /usr/local/lib/python3.11/dist-packages (from onnxruntime==1.14.1->chromadb)
Requirement already satisfied: sympy in /usr/local/lib/python3.11/dist-packages (from onnxruntime==1.14.1->chromadb)
Requirement already satisfied: distro<2,>=1.7.0 in /usr/local/lib/python3.11/dist-packages (from openai<2.0.0,>=1.68.2->langchain-openai)
Requirement already satisfied: jiter<1,>=0.4.0 in /usr/local/lib/python3.11/dist-packages (from openai<2.0.0,>=1.68.2->langchain-openai)
Requirement already satisfied: sniffio in /usr/local/lib/python3.11/dist-packages (from openai<2.0.0,>=1.68.2->langchain-openai)
Requirement already satisfied: deprecated==1.2.6 in /usr/local/lib/python3.11/dist-packages (from opentelemetry-api>=1.24.0->opentelemetry-exporter-otlp-proto-common)
Requirement already satisfied: importlib-metadata<8.7.0,>=6.0 in /usr/local/lib/python3.11/dist-packages (from opentelemetry-exporter-otlp-proto-common)
Requirement already satisfied: googleapis-common-protos<1.52 in /usr/local/lib/python3.11/dist-packages (from opentelemetry-exporter-otlp-proto-common)
Requirement already satisfied: opentelemetry-exporter-otlp-proto-common==1.31.1 in /usr/local/lib/python3.11/dist-packages (from opentelemetry-exporter-otlp-proto-common)
Requirement already satisfied: opentelemetry-protocol==1.31.1 in /usr/local/lib/python3.11/dist-packages (from opentelemetry-exporter-otlp-proto-common)
Requirement already satisfied: opentelemetry-instrumentation-asgi==0.52b1 in /usr/local/lib/python3.11/dist-packages (from opentelemetry-instrumentation-asgi)
Requirement already satisfied: opentelemetry-instrumentation==0.52b1 in /usr/local/lib/python3.11/dist-packages (from opentelemetry-instrumentation-asgi)
Requirement already satisfied: opentelemetry-semantic-conventions==0.52b1 in /usr/local/lib/python3.11/dist-packages (from opentelemetry-instrumentation-asgi)
Requirement already satisfied: opentelemetry-util-http==0.52b1 in /usr/local/lib/python3.11/dist-packages (from opentelemetry-instrumentation-asgi)
Requirement already satisfied: wrapt<2.0.0,>=1.0.0 in /usr/local/lib/python3.11/dist-packages (from opentelemetry-instrumentation-asgi)
Requirement already satisfied: asgiref<4.0 in /usr/local/lib/python3.11/dist-packages (from opentelemetry-instrumentation-asgi)
Requirement already satisfied: monotonic<1.5 in /usr/local/lib/python3.11/dist-packages (from posthog>=2.4.0->chromadb)
Requirement already satisfied: backoff>=1.10.0 in /usr/local/lib/python3.11/dist-packages (from posthog>=2.4.0->chromadb)
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.11/dist-packages (from pydantic>=1.9.0->pydantic-core)
Requirement already satisfied: pydantic-core==2.33.0 in /usr/local/lib/python3.11/dist-packages (from pydantic>=1.9.0->pydantic-core)
Requirement already satisfied: typing-inspection>=0.4.0 in /usr/local/lib/python3.11/dist-packages (from pydantic>=1.9.0->pydantic-core)
Requirement already satisfied: python-dotenv>=0.21.0 in /usr/local/lib/python3.11/dist-packages (from pydantic-settings)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.11/dist-packages (from requests<3,>=2.31.0->requests)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.11/dist-packages (from rich>=10.11.0->rich)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.11/dist-packages (from rich>=10.11.0->rich)
Requirement already satisfied: greenlet>=1 in /usr/local/lib/python3.11/dist-packages (from SQLAlchemy>=1.4.0->SQLAlchemy)
Requirement already satisfied: regex>=2022.1.18 in /usr/local/lib/python3.11/dist-packages (from tiktoken<1,>=0.7->tiktoken)
Requirement already satisfied: huggingface-hub<1.0,>=0.16.4 in /usr/local/lib/python3.11/dist-packages (from tokenizers)
Requirement already satisfied: click>=8.0.0 in /usr/local/lib/python3.11/dist-packages (from typer>=0.9.0->typer)
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.11/dist-packages (from typer>=0.9.0->typer)
Requirement already satisfied: httpx>=0.6.3 in /usr/local/lib/python3.11/dist-packages (from uvicorn[standard]>=0.19.0->uvicorn)
Requirement already satisfied: uvloop!=0.15.0,!0.15.1,>=0.14.0 in /usr/local/lib/python3.11/dist-packages (from uvicorn[standard]>=0.19.0->uvicorn)
Requirement already satisfied: watchfiles>=0.13 in /usr/local/lib/python3.11/dist-packages (from uvicorn[standard]>=0.19.0->uvicorn)
Requirement already satisfied: websockets>=10.4 in /usr/local/lib/python3.11/dist-packages (from uvicorn[standard]>=0.19.0->uvicorn)
Requirement already satisfied: cachetools<6.0,>=2.0.0 in /usr/local/lib/python3.11/dist-packages (from google-auth==1.0.1->google-auth)
Requirement already satisfied: pyasn1-modules>=0.2.1 in /usr/local/lib/python3.11/dist-packages (from google-auth==1.0.1->google-auth)
Requirement already satisfied: rsa<5,>=3.1.4 in /usr/local/lib/python3.11/dist-packages (from google-auth==1.0.1->google-auth)
Requirement already satisfied: filelock in /usr/local/lib/python3.11/dist-packages (from huggingface-hub<1.0,>=0.16.4->huggingface-hub)
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.11/dist-packages (from huggingface-hub<1.0,>=0.16.4->huggingface-hub)
Requirement already satisfied: zipp>=3.20 in /usr/local/lib/python3.11/dist-packages (from importlib-metadata<8.7.0,>=6.0->importlib-metadata)
Requirement already satisfied: jsonpointer>=1.9 in /usr/local/lib/python3.11/dist-packages (from jsonpatch<2.0,>=1.33->requests-toolbelt)
Requirement already satisfied: mdurl<0.1 in /usr/local/lib/python3.11/dist-packages (from markdown-it-py>=2.2.0->markdown-it-py)
Requirement already satisfied: mypy_extensions>=0.3.0 in /usr/local/lib/python3.11/dist-packages (from typing-inspection>=0.4.0->typing-inspection)
Requirement already satisfied: humanfriendly>=9.1 in /usr/local/lib/python3.11/dist-packages (from coloredlogs>=0.6.2->coloredlogs)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.11/dist-packages (from sympy>=1.12.0->sympy)
Requirement already satisfied: pyasn1<0.7.0,>=0.6.1 in /usr/local/lib/python3.11/dist-packages (from pyasn1-modules>=0.2.1->pyasn1-modules)
```

```
from google.colab import userdata
import os
os.environ['OPENAI_API_KEY'] = userdata.get('openAIYtKey')
```

```
from langchain_community.document_loaders import WebBaseLoader
```

```
loader = WebBaseLoader(web_paths=["https://www.educosys.com/course/genai"])
```

```
docs = loader.load()
print(docs)
```

```
WARNING:langchain_community.utils.user_agent:USER_AGENT environment variable not set, consider setting it to identify your application
[Document(metadata={'source': 'https://www.educosys.com/course/genai', 'title': 'Hands-on Generative AI Course', 'description': 'A hands-on course for learning Generative AI using OpenAI GPT-4o and LangChain. The course covers the fundamentals of Generative AI, including prompt engineering, model evaluation, and deployment. It also includes practical exercises and projects to help you build your own Generative AI applications.'})]
```

```
from langchain.text_splitter import RecursiveCharacterTextSplitter
```

```
text_splitter = RecursiveCharacterTextSplitter(chunk_size = 1000, chunk_overlap = 200)
splits = text_splitter.split_documents(docs)
```

```
print(splits[0])
print(splits[1])
print(splits[2])
```

```

page_content='Hands-on Generative AI CourseCoursesBundle CoursesMentorFree ContentTestimonialsFAQLogin Signup Hands-on G
page_content='(VAEs) Probabilistic Data Generation Using VAEs Four Mini Projects using TensorFlow Metrics Visualization
page_content='Nodes, State, StateGraph, Workflows AI Agents Mini Project – Simple Q&A Application Using LangChain5Week 5

print(len(splits))

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from langchain.embeddings.openai import OpenAIEmbeddings
from langchain.vectorstores import Chroma

vectorstore = Chroma.from_documents(documents=splits, embedding=OpenAIEmbeddings())

<ipython-input-7-f24fa4352e95>:4: LangChainDeprecationWarning: The class `OpenAIEmbeddings` was deprecated in LangChain
vectorstore = Chroma.from_documents(documents=splits, embedding=OpenAIEmbeddings())

print(vectorstore._collection.count())

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print(vectorstore._collection.get())

{'ids': ['3a550eec-2fab-40a1-8b74-ea1a26fb34c2', '905a995d-6fd2-46f2-ae97-2cc010f00cde', 'f5109d13-0b7b-421e-885d-f558c2

print("\nCollection 1 - ", vectorstore._collection.get(ids=['28651d9a-ab51-41f8-ab83-e68285623c4e'], include=["embeddings",
print("\nCollection 2 - ", vectorstore._collection.get(ids=['054dee19-19ed-4574-bc51-511060fd707a'], include=["embeddings",
print("\nCollection 3 - ", vectorstore._collection.get(ids=['2fd71cb4-835a-43c5-b920-b7e1be51c450'], include=["embeddings",

Collection 1 - {'ids': [], 'embeddings': array([], dtype=float64), 'documents': [], 'uris': None, 'included': ['embeddi
Collection 2 - {'ids': [], 'embeddings': array([], dtype=float64), 'documents': [], 'uris': None, 'included': ['embeddi
Collection 3 - {'ids': [], 'embeddings': array([], dtype=float64), 'documents': [], 'uris': None, 'included': ['embeddi

retriever = vectorstore.as_retriever()

from langchain import hub
prompt = hub.pull("rlm/rag-prompt")

/usr/local/lib/python3.11/dist-packages/langsmith/client.py:278: LangSmithMissingAPIKeyWarning: API key must be provided
warnings.warn(

from langchain_openai import ChatOpenAI
llm = ChatOpenAI()

from langchain_core.runnables import RunnablePassthrough
from langchain_core.output_parsers import StrOutputParser

def format_docs(docs):
    return "\n".join(doc.page_content for doc in docs)

rag_chain = ({ "context" : retriever | format_docs, "question" : RunnablePassthrough() }
            | prompt
            | llm
            | StrOutputParser())

rag_chain.invoke("Are the recordings of the course available? For how long?")

'Yes, the recordings of the course are available for lifetime access.'

rag_chain.invoke("Are the testimonials for the course available? Name the student who have shared testimonials")

'Yes, testimonials for the course are available on the website. Students who have shared testimonials are Sahitya Raj,
Manika Kaushik, Ashish Upreti, and Sathish Krishna.'

rag_chain.invoke("Are the certificates for the course provided?")

'Yes, certificates are provided for the course.'

```

```
rag_chain.invoke("What all projects are covered in the course?")
```

↗ 'The projects covered in the course include implementing GANs to generate handwritten digits, training a VAE to generate faces using the CelebA dataset, building a transformer from scratch, and performing sentiment analysis using BERT. The course also covers topics such as attention mechanisms, transformers, fine-tuning techniques, and vector databases. It offers hands-on learning experiences and allows participants to work on mini and major projects to enhance their skill

```
from langchain_core.runnables import RunnableLambda
```

```
def print_prompt(prompt_text):
    print("Prompt - ", prompt_text)
    return prompt_text
```

```
rag_chain_with_print = ({ "context" : retriever | format_docs, "question" : RunnablePassthrough() }
                        | prompt
                        | RunnableLambda(print_prompt)
                        | llm
                        | StrOutputParser())
```

```
rag_chain_with_print.invoke("What all projects are covered in the course?")
```

↗ Prompt - messages=[HumanMessage(content="You are an assistant for question-answering tasks. Use the following pieces of 'The projects covered in the course include implementing a GAN to generate handwritten digits, training a VAE to generate faces using the CelebA dataset, code Transformer from scratch, and sentiment analysis using BERT. There are also mini-projects related to TensorFlow, TensorBoard, and various deep generative models. The course covers a range of topics related to generative AI and provides hands-on learning opportunities.'"])

Start coding or generate with AI.